

Dual-Head GraphSAGE Ensemble with Reconstruction and Discriminative Learning for Class-Imbalanced IoT Intrusion Detection

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Abstract— *The rapid proliferation of IoT networks has led to an increase in diverse cyberattacks accompanied by class imbalance and traffic distribution shifts. Existing GraphSAGE and Transformer-based Masked Autoencoder (MAE) models suffer from a structural limitation of relying solely on reconstruction error, achieving Macro F1 scores of only 0.435 and 0.502 on IoT datasets such as NF-BoT-IoT-v3-merged and NF-ToN-IoT-v3, respectively. In this paper, we analyze the root causes of these failures and propose a Dual-Head GraphSAGE ensemble architecture that combines reconstruction-based anomaly detection with discriminative learning. The proposed model consists of a skip connection that preserves original edge features, a MAE reconstruction head, and a classifier head trained with Focal BCE and Soft Macro F1 Loss. Experimental results show that the proposed model achieves Macro F1 scores of 0.922 and 0.909 on NF-BoT-IoT-v3-merged and NF-ToN-IoT-v3, respectively, while maintaining baseline-level performance (Macro F1 0.9998) on NF-UNSW-NB15-v3. These results demonstrate that the proposed architecture improves robustness in IoT-specific class-imbalanced environments while preserving general-purpose intrusion detection performance.*

Keywords— *Graph Neural Network, GraphSAGE, Intrusion Detection, IoT Security, Class Imbalance, Dual-Head Ensemble, Masked Autoencoder*

I. INTRODUCTION

The rapid proliferation of Internet of Things (IoT) devices has significantly increased the number of network-connected endpoints, making IoT networks a primary target for diverse cyberattacks including botnets, DDoS, scanning, and injection attacks. Since IoT devices typically operate under constrained computational resources and limited security capabilities, the importance of real-time network intrusion detection systems continues to grow.

Existing GraphSAGE and Transformer MAE-based intrusion detection models [2] rely on reconstruction error-based anomaly detection and achieve a Macro F1 of 0.974 on general-purpose datasets (NF-UNSW-NB15-v3). However, performance degrades substantially on IoT-specific datasets (NF-BoT-IoT-v3-merged, NF-ToN-IoT-v3), dropping to 0.435 and 0.502, respectively. This failure stems from two structural causes. First, in severely class-imbalanced

environments, the reconstruction error distributions of benign and malicious classes overlap, rendering threshold-based classification unstable. Second, the GraphSAGE neighborhood aggregation process dilutes original edge-level traffic features, degrading detection performance for subtle attack patterns.

To address these issues, this work proposes a Dual-Head GraphSAGE ensemble model that complementarily combines reconstruction-based anomaly scoring with discriminative class boundary learning. The main contributions of this paper are as follows:

1. We analyze the root causes of performance degradation in GraphSAGE+MAE-based models under class-imbalanced IoT environments.
2. We propose a Dual-Head GraphSAGE ensemble architecture with a skip connection that preserves original edge features.
3. We combine Focal BCE and Soft Macro F1 Loss to directly optimize Macro F1 under class-imbalanced conditions.
4. We validate the contribution of each component through ablation study and alpha sensitivity analysis.
5. We confirm performance preservation on NF-UNSW-NB15-v3 and evaluate result stability through multi-seed validation.

II. RELATED WORK

A. GNN-Based Network Intrusion Detection

Network traffic can be represented as a graph by modeling hosts, flows, and communication entities as nodes and edges. GNN-based approaches [4] are capable of capturing relational information such as repeated connections, anomalous communication paths, and attack propagation patterns, offering advantages over traditional machine learning methods for complex network traffic analysis. In particular, NetFlow-based flow records can be naturally converted into graph structures using source and destination information, making them well-suited for GNN model application [11].

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B. GraphSAGE-Based Representation Learning

GraphSAGE [3] is an inductive graph neural network that learns node representations by sampling and aggregating features from neighboring nodes. It can generate embeddings for nodes not observed during training, making it suitable for dynamic network environments. However, the neighborhood aggregation process can dilute fine-grained original edge features such as packet counts, byte counts, and flow duration. Since such features may be over-smoothed during aggregation, a mechanism to preserve original edge features is necessary when applying GraphSAGE to intrusion detection tasks.

C. Masked Autoencoder-Based Anomaly Detection

Masked Autoencoders [5] learn representations by masking a portion of input features and reconstructing the missing information in a self-supervised manner. In intrusion detection, reconstruction-based models assume that samples deviating from normal traffic patterns generate larger reconstruction errors. Models combining GraphSAGE and Transformer MAE [2] have demonstrated high performance on general-purpose intrusion detection datasets. However, in class-imbalanced IoT environments, the reconstruction error distributions of benign and malicious classes can overlap, making threshold-based classification unstable.

D. Handling Class Imbalance

Class imbalance is a common challenge in network intrusion detection datasets and is particularly severe in IoT environments. Under such conditions, accuracy can be misleading as models may achieve high accuracy by focusing on the majority class while failing to detect minority class samples. Macro F1 is widely used as an evaluation metric since it assigns equal importance to each class. Focal Loss [6] improves minority class learning by assigning larger gradients to hard or misclassified samples, while Soft Macro F1 Loss [7] approximates Macro F1 in a differentiable form to directly optimize class-balanced performance.

E. NetFlow-Based Intrusion Detection Datasets

This work uses NF-UNSW-NB15-v3 [8], NF-BoT-IoT-v3-merged [9], and NF-ToN-IoT-v3 [10]. NF-UNSW-NB15-v3 is a general-purpose network intrusion detection dataset, while NF-BoT-IoT-v3 and NF-ToN-IoT-v3 contain IoT-specific attack types including botnet, DoS, DDoS, reconnaissance, scanning, and injection [11]. Since IoT datasets exhibit traffic distributions and class imbalance characteristics that differ substantially from general-purpose datasets, models that operate robustly in both environments are needed.

F. Limitations of Reconstruction-Based GNN Intrusion Detection

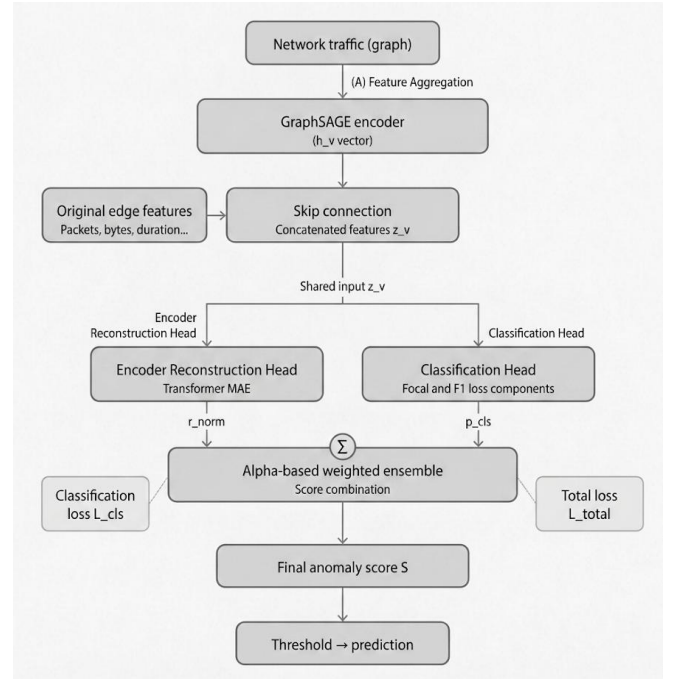
Existing GraphSAGE+MAE-based models [2] rely heavily on reconstruction error and are effective when benign and malicious samples produce clearly separated reconstruction error distributions. However, in class-imbalanced IoT environments, the reconstruction error distributions of the two classes can overlap, making threshold-based classification unreliable. Furthermore, GraphSAGE neighborhood aggregation dilutes original edge-level traffic features, degrading detection performance for attack patterns reflected in fine-grained flow statistics. Therefore, additional mechanisms are needed to preserve original flow information and learn explicit class boundaries.

III. PROPOSED METHOD

A. Overall Architecture

The proposed model consists of a GraphSAGE-based graph encoder, a skip connection to preserve original edge features, a Transformer MAE reconstruction head, and a classifier head. Input network traffic is converted into a graph structure, and the GraphSAGE encoder aggregates neighborhood information to generate embeddings. The encoder output is then concatenated with the original edge features and fed into two heads. The MAE reconstruction head generates an anomaly score based on reconstruction error, while the classifier head outputs the probability of malicious traffic. The final anomaly score is computed through a weighted ensemble of the two head outputs.

Fig.1 Overall architecture of the proposed Dual-Head GraphSAGE ensemble model.



B. Edge Feature Preservation via Skip Connection

While GraphSAGE can integrate graph structural information through neighborhood aggregation, fine-grained original edge features of individual flows may be diluted during this process. In network intrusion detection, original features such as packet counts, byte counts, flow duration, and protocol statistics play an important role in attack detection. Therefore, this work employs a skip connection that directly concatenates the GraphSAGE encoder output with the original edge features:

$$\mathbf{z}_v = [\mathbf{h}_v \parallel \mathbf{e}_v]$$

where $\mathbf{h}_v$ denotes the GraphSAGE encoder output, $\mathbf{e}_v$ denotes the original edge feature vector, and $\parallel$ denotes vector concatenation. The combined representation $\mathbf{z}_v$ is used as the shared input to both the MAE reconstruction head and the classifier head.

B. Dual-Head Architecture

The proposed model adopts a dual-head architecture consisting of a MAE reconstruction head and a classifier head. The MAE reconstruction head follows the Transformer MAE structure and is trained to reconstruct partially masked input features. It uses reconstruction error to measure how much an input sample deviates from normal patterns.

The classifier head uses label information to learn explicit decision boundaries between benign and malicious classes. Since reconstruction-based anomaly detection alone may not reliably identify benign classes in highly imbalanced IoT datasets, the classifier head complements the reconstruction head by improving discriminative performance.

C. Loss Function

The total loss consists of reconstruction loss and classification loss:

$$L_{total} = L_{rec} + \lambda L_{cls}$$

The classification loss is defined as a combination of Focal BCE and Soft Macro F1 Loss:

$$L_{cls} = L_{focal} + \beta L_{softF1}$$

Focal BCE assigns larger gradients to hard or misclassified samples to improve minority class learning. Soft Macro F1 Loss helps directly optimize Macro F1 and contributes to balanced performance between benign and malicious classes under class-imbalanced conditions.

D. Alpha-Based Weighted Ensemble

During inference, the final anomaly score is computed by combining the malicious probability from the classifier head and the reconstruction error from the MAE head:

$$\mathbf{S} = \alpha \cdot \mathbf{p}_{cls} + (1 - \alpha) \cdot \mathbf{r}_{norm}$$

where $\mathbf{p}_{cls}$ is the malicious class probability generated by the classifier head, $\mathbf{r}_{norm}$ is the normalized MAE reconstruction error, and α controls the relative contribution of the two heads. When $\alpha=0$, only reconstruction error is used; when $\alpha=1$, only the classifier probability is used.

IV. EXPERIMENTAL SETUP

A. Datasets

This work uses three NetFlow-based intrusion detection datasets. NF-BoT-IoT-v3-merged is an IoT intrusion detection dataset containing attacks such as botnet, DoS, DDoS, and reconnaissance. NF-ToN-IoT-v3 includes diverse IoT attack types including scanning, injection, and DoS. NF-UNSW-NB15-v3 is used as a general-purpose network intrusion detection dataset to verify that the proposed model maintains performance beyond IoT-specific environments.

B. Evaluation Metrics

Since accuracy can be biased toward the majority class in class-imbalanced environments, this work uses Macro F1 as the primary evaluation metric. Weighted F1, Benign F1, Malicious F1, PR-AUC, and ROC-AUC are also used for comprehensive performance analysis. Benign F1 is particularly important in IoT datasets where benign classes may be underrepresented.

C. Experimental Configuration

This work conducts the following six experiments:

1. main IoT performance comparison
2. performance preservation on NF-UNSW-NB15-v3,
3. ablation study
4. alpha sensitivity analysis
5. imbalance distribution evaluation
6. multi-seed validation.

D. Implementation Details and Evaluation Protocol

A consistent evaluation protocol was applied across all experiments to ensure fair comparison between the baseline and the proposed model. Each dataset was split into training, validation, and test sets, and the validation set was used to select the optimal model based on Macro F1. Reconstruction errors were normalized prior to weighted ensemble to reduce the effect of scale differences across datasets. Multi-seed validation was conducted using seeds 42, 77, 100, 123, and 2026, and results are reported as mean and standard deviation of Macro F1, Benign F1, and Malicious F1. All experiments were conducted on a system equipped with an AMD Ryzen 7 7800X3D CPU, 64GB RAM, and an NVIDIA GeForce RTX 5070 Ti GPU (16GB VRAM) with CUDA 12.9.

Table I. Experimental Configuration for IoT Datasets

Model	Dataset	Macro F1	Benign F1	Malicious F1	PR-AUC	ROC-AUC
Baseline	NF-BoT-IoT-v3-merged	0.435	-	-	0.990	-
	NF-ToN-IoT-v3	0.502	-	-	0.428	-
Proposed	NF-BoT-IoT-v3-merged	0.922	0.882	0.962	0.985	0.965
	NF-ToN-IoT-v3	0.909	0.923	0.895	0.954	0.961

V. RESULTS AND ANALYSIS

A. Main Performance Comparison on IoT Datasets

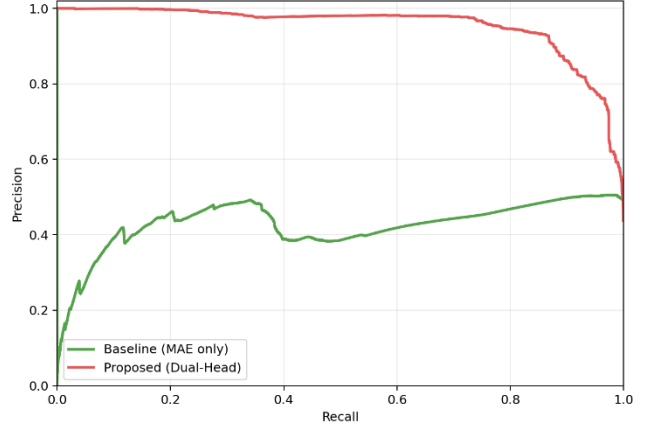
The proposed model achieved higher Macro F1 scores than the existing GraphSAGE+MAE baseline on both IoT datasets. On NF-BoT-IoT-v3-merged, the Macro F1 score improved from 0.435 to 0.922. On NF-ToN-IoT-v3, the Macro F1 score improved from 0.502 to 0.909.

Table II. Main Performance Comparison on IoT Datasets

Hyperparameter	BoT-IoT-merged	ToN-IoT
GraphSAGE hidden dim	128	128
GraphSAGE layers	1	1
Transformer hidden dimension	128	512
Transformer attention dimension	128	128
Masking ratio	0.25	0.20
Learning rate	5e-4	2e-4
Optimizer	Adam	Adam
Batch size	4096	4096
Epochs	300	300
λ (classification loss weight)	0.3	0.2
β (Soft Macro F1 loss weight)	0.7	0.7
Dropout	0.3	0.3
gamma (Focal Loss)	2.0	3.0
selected ensemble_alpha	0.75	0.50

These results demonstrate that the proposed model provides more stable discriminative performance than the reconstruction-based baseline in class-imbalanced IoT environments. Fig. 2 presents the PR curve comparison between the Baseline and Proposed models on NF-ToN-IoT-v3, visually confirming that the proposed model maintains higher precision across most of the recall range.

Fig.2 PR curve comparison on NF-ToN-IoT-v3 (Baseline vs. Proposed).



B. Performance Preservation on NF-UNSW-NB15-v3

To verify that the proposed model does not overfit to IoT-specific datasets, it was evaluated on NF-UNSW-NB15-v3.

Table III. Performance Preservation on NF-UNSW-NB15-v3

Model	Macro F1	Weighted F1	Benign F1	Malicious F1	PR-AUC	ROC-AUC
Baseline	0.9999	-	-	-	0.9999	-
Proposed	0.9998	0.9999	1.0000	0.9995	1.0000	1.0000

The Macro F1 difference between the baseline and the proposed model is negligible. This indicates that the proposed model maintains baseline-level performance on the general-purpose network intrusion detection dataset while improving performance on IoT datasets.

C. Ablation Study

An ablation study was conducted to examine the contribution of each component of the proposed model.

Table IV. Ablation Study Results

Variant	Macro F1	Benign F1	Malicious F1
<i>NF-BoT-IoT-v3-merged</i>			
baseline_mae	0.4977	-	-
skip_only	0.6038	0.2310	0.9765
classifier_bce	0.7957	0.5942	0.9972
classifier_soft_f1	0.8686	0.7394	0.9979
full_dual_head	0.9220	0.8820	0.9620
<i>NF-ToN-IoT-v3</i>			
baseline_mae	0.5040	-	-
skip_only	0.6609	0.7529	0.5689
classifier_bce	0.8879	0.8991	0.8767
classifier_soft_f1	0.8878	0.8991	0.8765
full_dual_head	0.9090	0.9230	0.8950

On NF-BoT-IoT-v3-merged, the classifier_soft_f1 variant showed the strongest performance among the single-classifier variants, confirming the contribution of Soft Macro F1 Loss to benign class recovery. After alpha calibration, the full_dual_head model achieved the best overall Macro F1 on both NF-BoT-IoT-v3-merged and NF-ToN-IoT-v3, demonstrating that the combination of reconstruction and classifier heads provides complementary benefits depending on dataset characteristics.

D. Alpha Sensitivity Analysis

The effect of the alpha value in the dual-head ensemble was analyzed.

Table V. Alpha Sensitivity Results

Dataset	Best Alpha	Best Macro F1	Interpretation
NF-BoT-IoT-v3-merged	0.75	0.922	Classifier-oriented ensemble is more effective
NF-ToN-IoT-v3	0.50	0.909	Balanced combination of two heads is more effective

Results indicate that dataset-specific alpha calibration improves the effectiveness of the dual-head ensemble. The best alpha values were 0.75 for NF-BoT-IoT-v3-merged and 0.50 for NF-ToN-IoT-v3, suggesting that classifier probabilities and MAE reconstruction errors provide complementary information with different relative contributions depending on the dataset.

E. Imbalance Distribution Evaluation

To examine the stability of the proposed model under different class distributions, evaluation was conducted under balanced, original, 1:5, and 1:10 settings.

Table VI. Imbalance Distribution Results

Dataset	Balanced	Original	1:5	1:10
NF-BoT-IoT-v3-merged	0.928	0.922	0.914	0.907
NF-ToN-IoT-v3	0.913	0.909	0.902	0.896

Macro F1 scores decreased slightly as the evaluation distribution became more imbalanced, but the overall degradation remained limited. This suggests that the proposed model maintains relatively robust performance under varying class distributions.

F. Multi-Seed Validation

To reduce dependence on a single random seed, repeated experiments were conducted using five seeds: 42, 77, 100, 123, and 2026.

Table VII. Multi-Seed Validation Results

Dataset	Macro F1 Mean $\pm$ Std	Benign F1 Mean $\pm$ Std	Malicious F1 Mean $\pm$ Std
NF-BoT-IoT-v3-merged	0.914 $\pm$ 0.018	0.866 $\pm$ 0.031	0.962 $\pm$ 0.006
NF-ToN-IoT-v3	0.905 $\pm$ 0.006	0.919 $\pm$ 0.005	0.891 $\pm$ 0.008
NF-UNSW-NB15-v3	0.9997 $\pm$ 0.0002	1.0000 $\pm$ 0.0000	0.9994 $\pm$ 0.0003

NF-ToN-IoT-v3 and NF-UNSW-NB15-v3 showed stable performance with low standard deviation. NF-BoT-IoT-v3-merged still showed some remaining variance in Benign F1, indicating that threshold calibration and sampler configuration can affect benign class performance stability.

G. Summary of Experimental Results

The experimental results demonstrate that the proposed model improves IoT intrusion detection performance from multiple perspectives. The main performance comparison shows that the proposed dual-head GraphSAGE model substantially improves Macro F1 on both NF-BoT-IoT-v3-merged and NF-ToN-IoT-v3, indicating that reconstruction-based anomaly detection alone is insufficient in class-imbalanced IoT environments and that explicit discriminative learning is necessary. The NF-UNSW-NB15-v3 experiment confirms that the proposed model does not sacrifice performance on general-purpose intrusion detection datasets. The ablation study shows that Soft Macro F1 Loss improves benign class recovery, while the calibrated full_dual_head model provides the strongest overall Macro F1. Alpha sensitivity analysis supports the complementary relationship between classifier probabilities and reconstruction errors, and multi-seed validation further confirms the overall stability of the proposed approach.

VI. DISCUSSION

The experimental results indicate that the performance improvements of the proposed model primarily stem from three factors. First, the skip connection preserves original edge features that may be diluted during GraphSAGE neighborhood aggregation. Second, the classifier head learns explicit decision boundaries between benign and malicious classes, which is difficult to achieve through reconstruction error alone. Third, Soft Macro F1 Loss directly optimizes per-class F1 performance and contributes to recovering benign class detection performance in class-imbalanced IoT environments.

The NF-UNSW-NB15-v3 experiment demonstrates that the proposed model maintains comparable performance to the baseline on general-purpose network intrusion detection datasets, indicating that the proposed architecture is not limited to IoT datasets and can operate stably in general network environments. Furthermore, alpha sensitivity analysis confirms that reconstruction-based anomaly detection and classification-based discriminative learning are

complementary, with dataset-specific alpha calibration improving the final ensemble performance.

VII. CONCLUSION

This work analyzed the root causes of performance degradation in existing GraphSAGE+MAE-based intrusion detection models under class-imbalanced IoT environments and proposed a dual-head GraphSAGE ensemble architecture. The proposed model combines skip connections for original edge feature preservation, a MAE reconstruction head, a classifier head, Focal BCE, and Soft Macro F1 Loss to substantially improve Macro F1 on IoT datasets while maintaining baseline-level performance on NF-UNSW-NB15-v3.

The ablation study showed that the classifier head and Soft Macro F1 Loss play a critical role in benign class recovery, and alpha sensitivity analysis demonstrated that a dataset-specific combination of reconstruction-based and classification-based scores is more effective than using either head alone.

Future work will focus on four directions. First, an adaptive alpha adjustment mechanism will be introduced to automatically determine the optimal alpha per dataset. Second, validation-based dynamic threshold calibration will be applied to improve benign class detection stability. Third, the architecture will be extended to temporal graph learning to capture time-varying traffic patterns. Fourth, heterogeneous graph modeling will be explored to reflect IoT device type and protocol heterogeneity.

VIII. ACKNOWLEDGMENT

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